Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Why is modulation essential in Communication Systems?	[2]
b)	Why are decibels used to express signal strength and power ratios?	[3]
c)	Give the Signal Space representation of QPSK.	[2]
d)	What is the bandwidth requirement of a PCM system?	[3]
e)	Define network fundamentals.	[2]
f)	What is the primary function of a pager in a paging system?	[3]
g)	How many satellites are typically required for a GPS receiver to determine its three-dimensional position accurately?	[2]
h)	What is the principle of total internal reflection, and how is it used in optical fibers to guide light?	[3]
i)	What is the role of WCDMA in enabling high-speed data transfer on mobile networks?	[2]
j)	What does UWB stand for, and how does it achieve high data transfer rates?	[3]

PART – B

(50 Marks)

2.a)	Explain the concept of decibels in communication systems.	
b)	Discuss the concept of gain in the context of antennas and amplifiers, and its role in Communication systems.	[5+5]

OR

3.a)	Discuss the electromagnetic spectrum and its significance in modern communication and technology.	
b)	Design a communication system for interplanetary exploration, considering the need for modulation, frequency translation, gain, and attenuation control.	[5+5]

4.a)	Explain quantization error and write an expression for maximum SNR in PCM system that uses linear quantization.	
b)	With a neat sketch, explain the modulation and detection of Differential Phase shift Keying.	[5+5]

OR

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5. Evaluate the spectral efficiency, error performance, and bandwidth utilization of ASK, FSK, PSK, and QPSK, and determine the optimal modulation scheme for different communication scenarios. [10]

6.a) Explain the fundamental principles behind the operation of telephone systems, including analog and digital telephony.

b) Discuss the various components of a paging system, and how they work together to deliver messages. [5+5]

OR

7.a) Explain the advantages of VoIP over traditional telephone systems, including cost savings, scalability, and integration with other digital services.

b) Discuss the function of LAN hardware devices such as network interface cards and routers in the context of local area networks. [5+5]

8.a) Discuss the working principle of satellite communication systems and how they facilitate long-distance communication across the globe? Describe the process of uplink and downlink in satellite communication and the role of transponders.

b) Analyze the concept of Wavelength Division Multiplexing (WDM) and how it enables multiple signals to be transmitted simultaneously through a single optical fiber? [5+5]

OR

9.a) Explain the working principle of Global Positioning Systems (GPS) and how satellites are utilized to provide accurate positioning and navigation services.

b) Discuss the structure and materials used in fiber-optic cables, including the core, cladding, and protective jacket. [5+5]

10.a) Explain the concept of Radio-Frequency Identification technology and how it enables contactless data transmission between RFID tags and readers.

b) Discuss the history and development of the Advanced Mobile Phone System (AMPS) as one of the earliest analog cellular telephone systems. [5+5]

OR

11.a) Discuss the role of CDMA in the development of third-generation (3G) mobile technologies and how it has contributed to the evolution of high-speed data services on mobile networks.

b) Explain the working principle of WiMAX as a wireless broadband technology. [5+5]

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